"Not just devices, but a bridge to independence."
— Orthotics & Prosthetics

Function
Mobility
Independence
Better Tomorrow

ORTHOTICS & PROSTHETICS

RESTORING MOVEMENT • REBUILDING LIVES

LEARN | UNDERSTAND | APPLY | HEAL



WHAT IS O&P?

Orthotics & Prosthetics (O&P) is the branch of rehabilitation that deals with the design, fabrication and fitting of orthotic devices (braces & supports) and **prosthetic limbs** to improve function, mobility and quality of life.



"Small Devices... Big Changes"

IMPORTANCE OF O&P

- Restores function and mobility
- Prevents deformities and complications
- Improves balance, gait and posture
- Enhances independence in daily life
- Supports return to work, school and sports

"To give someone their freedom back – that's the goal."

BRANCHES OF O&P

ORTHOTICS
(Braces & Supports)

- AFO (Ankle Foot Orthosis)
- KAFO (Knee Ankle Foot Orthosis)
- TLSO (Thoraco Lumbo Sacral Orthosis)
- Wrist / Hand Splints
- Custom Orthoses

PROSTHETICS
(Artificial Limbs)

- Upper Limb Prosthesis
- Lower Limb Prosthesis
- Myoelectric Prosthesis
- Cosmetic Prosthesis
- Modular Prosthesis

KEY CONCEPTS

- Biomechanics
- Fit & Alignment
- Gait Analysis
- Material Science
- Socket Design
- Componentry
- Patient Assessment
- Functional Training
- Maintenance & Follow-up
- Interdisciplinary Teamwork
- Customization
- Patient Education

CLINICAL CONDITIONS

- Amputations
- Neurological Disorders (Stroke, SCI, CP, etc.)
- Musculoskeletal Conditions (Arthritis, Fractures, Deformities)
- Orthopaedic Injuries
- Cerebral Palsy
- Spinal Cord Injury
- Rheumatoid Arthritis
- Diabetic Foot

Same person... New possibilities.





REMEMBER

- Individualized approach.
- Patient education.
- Function over form.
- Better mobility = Better life.

Move Better • Live Better

NAME : _____

SUBJECT : ORTHOTICS & PROSTHETICS

YEAR : _____

Future Rehab Specialist

SAFE | EFFECTIVE | ETHICAL | EVIDENCE BASED | PATIENT CENTERED

MEDICOS MINDS

HIGH-YIELD BPT STUDY GUIDE & EXAM COMPANION

Orthotics & Prosthetics

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OFFICIAL DIGITAL STUDY GUIDE VERIFICATION

Format: High-Yield Digital PDF | Standard Price: Rs. 59

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STUDY GUIDE DETAILS & CONTENTS:

Comprehensive high-yield notes designed for Bachelor of Physiotherapy (BPT), MBBS, BDS, and allied healthcare students. Covers fundamental anatomy, clinical assessments, pathological conditions, evidence-based physical therapy rehabilitation protocols, and exam-oriented high-yield review.

KEY HIGHLIGHTS:

- Exam-Focused: Concise points curated from leading medical textbooks and clinical references.
- Visual Diagrams: High-yield diagrams and tables for quick revision.
- Mobile & Tablet Optimized: Readable on phones, tablets, iPads, and laptops.
- Official Medicos Minds Release: Verified by Dr. MD Kamran Alam.

Hip - Knee - Ankle - Foot orthosis:

HKAFO is an orthosis whose components stabilize and lock the Hip, knee & Ankle



Parts:

* pelvic Band
* Hip joints & locks
* Ischial band
* Thigh Band
* knee cap & knee joint
* Calf band
* Metal uprights
* Boot
* Stirrup

Reciprocal Gait orthosis:

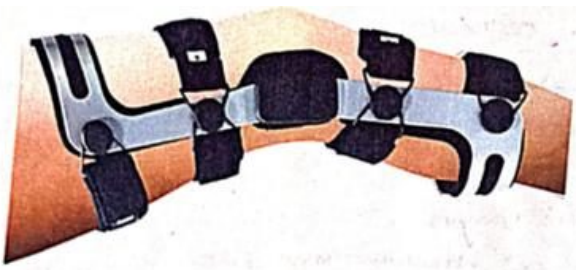
A Reciprocal Gait orthosis is an HKAFO that uses a mechanical system that connects the two sides of brace by
- Isocentric Rod
- Double cable
- Single push/pull cable system

- Components:

* AFO
* Knee joint
* uprights
* Thigh cuff
* Hip joint
* pelvic Band
* Cables
* Thoracic straps

Indications:

* Commonly used in cases of spinal cord injury, spina bifida.
* It combines flexion of one hip with extension of the opposite hip.
* The flexion power of one hip is utilised to extend the opposite hip



Indications:

* knee hyperextension
* Genu recurvatum

2) Donjoy Knee Brace:

Donjoy knee brace to suit all activity levels from everyday use to professional sports people



- Indications:

* Stability problems
  eg: ligament injuries
* use after knee surgery
* Arthritis sufferers
  → Reduce pain, improve function & prevent surgery
* prevent injury:-
  many sports professionals wear Donjoy knee braces → to reduce the risk of injury

3) Extension Knee orthosis:

* It consists of two long metal uprights pivoting thigh & calf cuffs
* It is used to protect knee against mediolateral forces.



THANK YOU FOR YOUR PURCHASE!

We wish you the very best in your medical and physiotherapy examinations.

For any questions, supplementary questions, or study support, reach out to us:

SUPPORT & COMMUNITY:

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